LESSON 10 · MISSION GUIDE

The War Table

groupby is the PivotTable, merge is XLOOKUP: 400 quests become a plan.

From **Captain Siv, Quest Master**
To **Jen, Keeper of the Guild Ledger**
By **raven, from the War Table**

Keeper, the War Table is the big map where I decide who goes where. On it: eleven PivotTables, pinned with daggers. Last spring one pointed at the wrong tab and I sent four Bards after a Troll. They came back. They have not stopped singing about it.

Orrin says pandas does a PivotTable in one line (`groupby`) and an XLOOKUP in one line (`merge`). If that's true, I want the whole table rebuilt from the ledger: which regions, which monsters, which notices are worth it. Yes or no answers, please. Dalvric's version once answered "mayhap". — S.

What you'll be able to do after today

- [ ] Count things by category with `value_counts()`
- [ ] Build a **PivotTable** with `groupby`: one or two row fields, any summary (mean, median, count, sum)
- [ ] Build the same with `pivot_table`, rows down, columns across, grand totals
- [ ] **XLOOKUP** a second table in with `merge`, and spot the rows it didn't find (NaN)
- [ ] Find and fill **missing values**: `.isna()`, `.fillna()`
- [ ] Cut values into **bands** with `qcut` and cross-tabulate them

Why this matters

- **Every summary table you've built in Excel is a groupby.** After today it's one line, and still right next month.
- **Joining two tables is most of real data work.** Two lists never agree; `merge` shows you exactly where.

Four big ideas

1. groupby = PivotTable	`groupby("region")["reward"].mean()`: row field, values, summary. Two row fields: a list.
2. unstack / pivot_table	`.unstack()` turns a row field into columns. `pivot_table` is the PivotTable dialog with names.
3. merge = XLOOKUP for whole tables	`merge(regions, on="region", how="left")`. Left keeps every quest (NaN if no match); inner silently drops the rest.
4. NaN is "no value"	A failed lookup. `.isna().sum()` counts them, `.fillna(x)` replaces them.

Three things that will bite you

- `groupby("a", "b")` is not two keys. `groupby(["a", "b"])` is.
- `how="inner"` loses rows without a word. Print `len()` before and after every merge.
- `.fillna(...)` gives you a new column; it doesn't change the old one unless you assign it back.

How today works

15 MIN
Lecture
Lars rebuilds the War Table's eleven PivotTables, live

60 MIN
Exercise sheet
Seven ravens: the War Table, region by region

15 MIN
Pages from the Grimoire, page 10
"war_table.py", found under the map

The ravens

#	FROM	RAVEN	YOU'LL USE
1	Siv	Counting heads	`value_counts()`, `normalize=True`
2	The Queen	The PivotTable	`groupby().mean()` `.agg([...])` `.idxmax()`
3	Siv	Two row fields	`groupby([a, b])`, `.unstack()`
4	The Queen	pivot_table, the real one	`pivot_table(..., margins=True)`
5	Orrin	XLOOKUP	`merge()`, `.isna()` `.fillna()`
6	Pell	Reward bands	`pd.qcut()`, `pd.crosstab()`
7 ⭐	Siv	Who goes where	two merges, named `.agg()`, `.loc[row, col]`

You're done when...

- [ ] `my_work/` has `ex1` to `ex6` (and `ex7` if the tavern rumour needs settling), each printing the expected output
- [ ] Dalvric's war_table runs with all 400 quests and the right percentages, one commit per bug
- [ ] Everything is pushed to `lesson-10` and you've opened a pull request

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Next lesson: The Vault Beneath. Under the guild hall is a vault, and in it, a database. Orrin has the key. You'll ask it questions in SQL, the language every data team on earth speaks, and it will answer in rows.